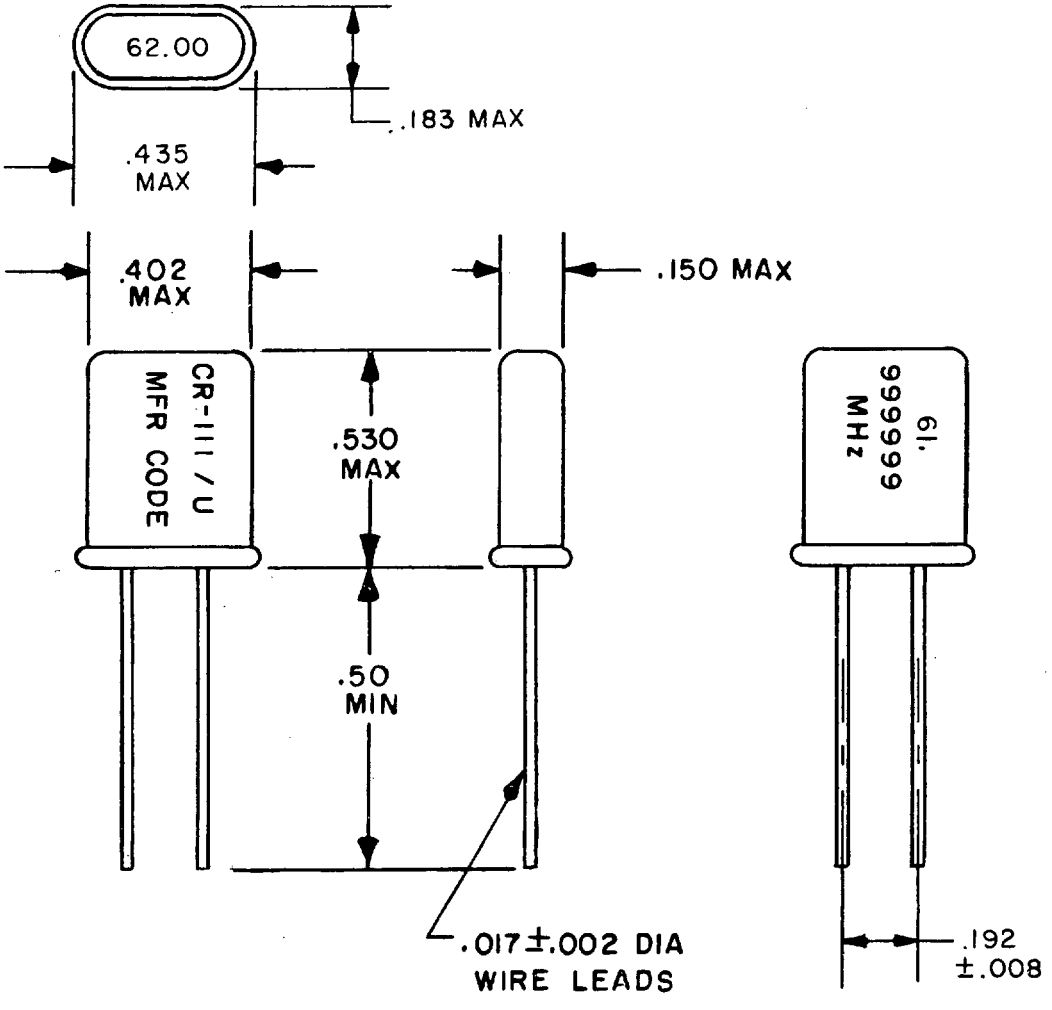
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-111/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 17 to 62 MHz; third mechanical overtone; noncontrolled; series resonance.



INCHES	MM
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (0.97 mm) in height and not less than .027 (0.69 mm) in width.

FIGURE 1. Crystal unit - CR-111/U.

Ⓒ denotes changes

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	17 to 62 MHz, inclusive
Capacitance, shunt	4 pF, maximum
Mode of oscillation	Third mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 ±0.4 mW
Calibration values:	
Equivalent resistance	30 ohms
Resistor voltage drop	.25 volts
Reference temperature	30°C ±1°C
Operating temperature range	-55°C to +105°C, inclusive
(C) Frequency tolerances:	
Operating temperature range	±50 ppm
Reference temperature	±15 ppm
Equivalent resistance	40 ohms, maximum
Unwanted modes 1/	Method II
Shock: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Vibration: 3/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Aging:	
Frequency change permitted	±5 ppm

1/ Unwanted modes: The minimum unwanted mode effective resistance shall be at least two times that of the main mode, or 40 ohms, whichever is greater.

2/ MIL-STD-202, method 202, 18 impact shocks of 5 g's each, time duration 11 ±1 milliseconds. Three shocks shall be applied in each direction along each of three mutually perpendicular planes.

3/ MIL-STD-202, method 204, test condition A, 0.03 inch double amplitude or 5 g's whichever is less.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11
DLA - ES

User activities:
Air Force - 17

Agent:
DLA - ES

Preparing activity:
Army - ER

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